

Frame It as Cases: Work That Speaks for Itself

FlyRank AI Internship

Week 2 – Assignment 2

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Voice Card

Simple, honest, direct, technical, human, concise.

Project 1 – Build Your First CRUD API

Problem

Before this assignment, I had very little understanding of how backend development worked. I knew that websites send and receive data, but I didn't know what happens after a client sends a request. The goal of this project was to understand how a backend receives requests, processes them, and returns the correct response using a CRUD API.

What I Did

I built a Task Management CRUD API using FastAPI. The API allows users to create, read, update, and delete tasks. While building it, I learned how different HTTP methods work, how routes are created, how request validation is handled using Pydantic, and how proper HTTP responses and status codes are returned.

The coding itself wasn't the hardest part. My biggest challenge was understanding when to use different FastAPI features, such as what parameters belong inside route decorators, which libraries need to be imported, and how request models work. I also learned an important lesson about how the return statement behaves inside loops while fixing an issue in my code. Another challenge was working with the terminal. I learned how to activate a virtual environment, run the FastAPI server, and test the API using Swagger UI.

Outcome

By the end of the assignment, I was able to build a working CRUD API from scratch. I now understand the basic request-response flow between a client and a backend application and can build small backend projects such as a task manager.

If I rebuild this project in the future, I would store the data in a database instead of a Python list and organize the code into a cleaner project structure.

Project 2 – BinarySpace: Autonomous Constellation Manager

Problem

During the National Space Hackathon 2026, our team wanted to build a system that could help monitor satellite constellations and reduce the risk of collisions. As the number of satellites and space debris increases, manually tracking and managing them becomes difficult. The challenge was to design a backend system that could process satellite information and support collision prediction.

What I Did

I worked on the backend part of the project. I learned how satellite data is received, processed, and prepared for the frontend. While building the backend, I worked with FastAPI and explored how orbital data can be handled using APIs and scientific Python libraries. One of the biggest challenges was understanding how different backend components communicate with each other and how live satellite data flows through the system. Working on this project also introduced me to concepts that I had never worked with before, such as orbital propagation and collision prediction.

Outcome

This project gave me my first experience of building a backend for a real-world problem instead of a practice application. I also learned how software development works in a team environment, where different members are responsible for different parts of the project. If I work on this project again, I would improve the backend architecture, add better data storage, and make the collision prediction system more efficient.

Project 3 – Redrob AI

Problem

Recruiters often spend a lot of time manually reviewing resumes before selecting suitable candidates. Our goal was to build a system that could automatically compare resumes with a job description and rank candidates based on their relevance.

What I Did

I worked on developing an AI-powered candidate ranking system. During the project, I learned how embeddings, semantic search, and similarity matching can be combined to compare resumes with job descriptions. One challenge was understanding how different AI components fit together instead of simply writing code. I also learned how backend services interact with AI models to process user requests and return meaningful results.

Outcome

This project introduced me to practical AI application development and showed me how AI can solve real recruitment problems. It also improved my understanding of integrating AI models into backend applications. If I rebuild this project, I would improve the ranking accuracy, support larger datasets, and make the user interface more interactive.

Project 4 – Urban Heat Island Mapping

Problem

Cities often experience higher temperatures than surrounding rural areas because of increasing urbanization. The objective of this project was to identify urban heat island regions using satellite imagery and geospatial analysis.

What I Did

I worked with Google Earth Engine to process satellite imagery and calculate different indices such as Land Surface Temperature (LST), NDVI, and NDBI. Before starting this project, I had very little experience with geospatial analysis. One of the biggest challenges was understanding satellite datasets, selecting suitable imagery, and interpreting the generated results. Throughout the project, I learned how satellite data can be used to study environmental conditions.

Outcome

This project introduced me to remote sensing and geospatial technologies. I learned how environmental data can be processed using Google Earth Engine and how different indices help explain urban temperature variations. If I revisit this project, I would perform a deeper analysis by comparing data across multiple years and adding more visualization features.

Bio

I am a Computer Science undergraduate at SVKM's NMIMS MPSTME with an interest in backend development, Java, and problem-solving. I enjoy learning by building projects because they help me understand how software works beyond theory. Through internships, hackathons, and personal projects, I have been developing my skills in backend development, AI applications, and geospatial technologies. My current focus is strengthening my backend fundamentals while preparing for software engineering roles.

Contact / Call to Action

If you'd like to discuss software development, backend engineering, AI, or hackathons, feel free to connect with me on LinkedIn or explore my projects on GitHub. I'm always open to learning from new experiences and collaborating on interesting ideas.

Before / After

Before

Developed an innovative AI-powered backend solution leveraging cutting-edge technologies to deliver scalable and efficient performance.

After

I worked on the backend of this project to understand how client requests are processed and how different components of a backend communicate with each other. Through this project, I learned new backend concepts by applying them to a real problem instead of only studying them in theory.